

COMPUTER SCIENCE TRIPOS Part IB 75%, Part II 50% – 2020 – Paper 7

5 Formal Models of Language (pjb48)

c, d, e are melodic elements. A birdsong is composed of a sequence of these elements as follows:

$c c d c c e c c d c c d c c e$

Grammar
Induction

- (a) (i) Using byte pair encoding induce a context-free grammar for this sequence of birdsong. Show your workings and state what you have decided to do in the case of a tie. [6 marks]

Answer: Several grammars are possible depending on what the candidate does with ties. Marks are awarded for: evidence of counting most frequent bigrams to select a pair to reduce to a non-terminal (1 mark); showing repeated application of this (1 mark); repeating until there are no pairs left with frequency > 1 (1 mark); stating what to do in the case of ties (1 mark); applying the previous point consistently (1 mark); finally producing a well formed CFG (1 mark).

Phrase-structure
derivation tree

- (ii) Draw the derivation tree that parses the birdsong using your induced grammar. [1 mark]

Answer: A parse tree consistent with the grammar the candidate produced.

Disadvantages of
induction method

- (iii) What are the shortcomings of this method of grammar induction for natural languages? [3 marks]

Answer: 3 points from the following: constituency might not be appropriate (1 mark); arbitrary decision is made for ties (1 mark); the string is assumed to be non-noisy (1 mark); extra-linguistic information, which could indicate constituency, is not taken into account (1 mark).

Entropy

- (b) Assuming that c, d and e are the only melodic elements available in the birdsong, and that the excerpt we are given is probabilistically representative of the birdsong in the wild, what is the average information produced per element? Provide relevant equations. [3 marks]

Answer:

$$H(X) = \sum_{x \in \mathcal{X}} p(x) \log_2 \left(\frac{1}{p(x)} \right) = - \sum_{x \in \mathcal{X}} p(x) \log_2 p(x)$$

equation (1 mark), calculation and substitution of relevant probabilities (e.g. $p(c) = 10/15 = 2/3$) (1 mark), an answer either written as full substitution into the equation or a final sum (1 mark).

2nd-order models
of entropy

- (c) In Part (b) we assumed a 1st-order model of the birdsong. What assumption does this make about the sequence of elements? [1 mark]

Answer: Elements are independent.

- (d) Consider a 2nd-order model of the birdsong and calculate the conditional entropy. Provide relevant equations. [4 marks]

Answer:

$$H(Y|X) = \sum_{x \in \mathcal{X}} p(x) H(Y|X=x) = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x,y) \log_2 p(y|x)$$

equation (1 mark), calculation and substitution of joint probabilities (e.g. $p(c,d) = 10/15 * 3/15$) (1 mark), calculation and substitution of conditional probabilities (e.g. $p(d|c) = 3/10$) (1 mark), an answer either written as full substitution into the equation or a final sum (1 mark).

entropy rate

- (e) How can we calculate the entropy rate of birdsong? Provide relevant equations. [2 marks]

Answer:

$$H_{\text{rate}}(L) = \lim_{n \rightarrow \infty} \frac{1}{n} H(X_1 \dots X_n)$$

Description of how to use this equation: for a stochastic process we can calculate entropy rate. The entropy of a language is the limit of the entropy rate of a sample of the language, as the sample gets longer and longer.
